Global Risk Consultants®

Global Risk Consultants Corp.
100 Walnut Avenue
Fifth Floor
Clark, New Jersey 07066
Phone: (732) 827-4400
Fax: (732) 827-4490

April 11, 2023

Najem Almamo
Planner & Scheduler
Cargill Incorporated
Starches and Sweeteners
Research & Development
84 Havenstraat
Vilvoorde, Belgium 1800

Re: Thermographic Survey - April 07, 2023
Research & Development - Vilvoorde, Belgium
File No. 495.1518
Client No. 10779

Dear Najem Almamo:

This report documents the results of my recent survey of the above-referenced location. Thermal imaging, ultrasonic listening and other technologies were utilized during this survey to locate abnormal electrical and mechanical conditions at your facility, which were indicative of impending failure. This report contains a list of equipment surveyed, documentation of detected recommendations and a summary of those recommendations.

Note that TÜV SÜD Global Risk Consultants augments our infrared thermographic inspection by utilizing state-of-the-art ultrasonic (airborne/structure-borne) listening equipment during the survey of your facility's and utility's electrical system. Our ultrasound technology allows us to listen for thermally undetectable corona and tracking problems in your facility's medium- or high-voltage equipment. It also allows us to listen for arcing in high-voltage switchgear that could not be opened for infrared inspection. Further, it is strategically utilized in low-voltage switchgear and/or transformers for anomalies mostly pertaining to mechanical looseness or contaminants.

As a result of this study of your plant's electrical power distribution and selected mechanical equipment, I report the detection of two (2) recommendations. The specifics of these recommendations are contained in the attached Recommendation Summary Report and detailed on an individual recommendation report page.

Also, please note that one (1) of the recommendations was appointed an FYI - "For Your Information" designation. This visually detected recommendation serve to bring to your attention conditions observed during the study, which may have the potential for negative impact on the integrity of the plant's operating systems. It documents previously unknown conditions of risk for reference and to assist in the mitigation of problem.

Of the two (2) recommendations detected during our last inspection, both were rechecked and found to be corrected.

As discussed during our closeout meeting:

1. An "Impact" rating is assigned to each detected recommendation. It is derived from our on-site discussions regarding the potential ramification each recommendation would have on business interruption and potential asset loss associated with the recommendations, should it reach a failure mode. To be specific, this is not a rating based on a recommendation's temperature. It is a rating that reflects the importance of the recommendation's potential damage to the plant's ongoing production and asset security. Impact ratings for detected faults are assigned after consultation with pertinent plant personnel during the on-site closeout meetings.

This report does not purport to set forth all hazards nor to indicate that other hazards do not exist. However, we urge that recommendations presented, if any, be given serious consideration. We would be pleased to discuss alternative solutions.

Draft Copy

Global Risk Consultants®

Global Risk Consultants Corp.
100 Walnut Avenue
Fifth Floor
Clark, New Jersey 07066
Phone: (732) 827-4400
Fax: (732) 827-4490

2. Both of the recommendations contained within this report presently fall within the realm of normal maintenance issues; for example: hot connections, defective components, deteriorated equipment, etc. As discussed, they require no special tools or repair technique to correct at this time. Each recommendation has been assigned a severity coding to assist you in the prioritization of repair.

Please note that your Corporate Risk Management Department requests that you review the recommendations and respond to each within 30 days of receiving your final report. GRC will send an email containing an electronic link within one to two business days of receiving your final report, which provides for an online response for each recommendation. Full instructions are also provided in the email. If there are no recommendations for this survey, you will not be receiving the response email.

Please convey my thanks to all your staff involved with this survey for the co-operation and assistance I received during this inspection. Should you have any questions regarding this report or should you require secondary a follow-up or emergency inspection; please contact me at yannick.ocke@tuvsud.com.

Sincerely,

Yannick Ocke

This report does not purport to set forth all hazards nor to indicate that other hazards do not exist. However, we urge that recommendations presented, if any, be given serious consideration. We would be pleased to discuss alternative solutions.

Draft Copy

Global Risk Consultants®

Global Risk Consultants Corp.
100 Walnut Avenue
Fifth Floor
Clark, New Jersey 07066
Phone: (732) 827-4400
Fax: (732) 827-4490

THERMOGRAPHIC SURVEY

Prepared for

Cargill Incorporated

Starches and Sweeteners
Research & Development
Vilvoorde, Belgium

File No. 495.1518
Client No. 10779

Accompanied By: Danny, Maintenance Technician

Exit Interview With: Najem Almamo, Planner & Scheduler

Thermographer: Yannick Ocke

April 07, 2023

This report does not purport to set forth all hazards nor to indicate that other hazards do not exist. However, we urge that recommendations presented, if any, be given serious consideration. We would be pleased to discuss alternative solutions.

Draft Copy

Global Risk Consultants®

INFRARED THERMOGRAPHY

Purpose:
Infrared Thermography is an integral part of any Preventive/Predictive Maintenance, Loss Prevention, and Risk Engineering Program. Utilizing infrared thermographic scanning on an annual basis for electrical and mechanical systems can improve the reliability of these systems and increase profitability. This is the primary reason for providing this service to you.

Background:
All objects emit electromagnetic radiation of a wavelength dependent on the object's temperature. The frequency of the radiation is inversely proportional to the temperature. In Infrared Thermography, this radiation is detected and measured by an infrared detector. The radiation is converted into electrical signals that can be displayed on a monitor. Permanent pictures (thermograms) of the thermal image as seen on the monitor are saved to compact flash disc. Using internal or external reference temperatures, the operator can calculate the temperature of the object being scanned. The thermal sensitivity of the state-of-the-art detectors used by Global Risk Consultants is 0.04°C at 30°C. Objects of an extremely wide range of temperatures can be scanned, as the state-of-the-art equipment features a temperature range from -40°C to +2000°C.

Infrared Thermography is an integral part of any Preventive/Predictive Maintenance, Loss Prevention, and Risk Engineering Program. Some of the advantages of infrared scanning are:

1. Abnormal thermal loading within operating/energized equipment may otherwise be unknown until a point of failure, without utilization of this technology. Thermographic imaging can help to identify and predict such failures so maintenance efforts can be executed prior to failure.
2. GRC's state-of-the-art infrared thermal imaging equipment utilized maintains precision accuracy (0.1 degree Celsius).
3. Discrete temperature measurements are provided to aid in determining the criticality of problems.
4. Measurements do not require contact with the object and can be made with the object under full load.
5. Infrared thermographic scanning is relatively quick; large foot-print areas can be scanned quickly because of the microsecond response of the technology and equipment utilized.
6. Safety; use of infrared thermal imaging equipment means measurements are generally made from a safe distance away from objects such as high voltage equipment, equipment at high temperatures, or equipment in a corrosive environment. However, the Thermographic Consultant still remains donned in appropriate Personal Protective Equipment (PPE) required by internationally recognized standards.

The following report lists the objects scanned during this survey and contains a Recommendation Summary Report. Also included are the probable cause and recommended corrective action for each finding with an abnormal temperature reading. For these objects, a photograph or printed image and a thermogram are provided, along with analysis data. A thermal severity and impact code classification system is utilized to capture thermal and non-thermal findings and their corresponding impact on the potential for business interruption.

Note: Temperature differences, even though small, can be serious, depending on the equipment, the load or voltage it is carrying, and the need for continuity of operation; therefore, the ultimate decision to repair such equipment must be made by qualified site personnel after evaluating the appropriate troubleshooting methods and techniques.

Global Risk Consultants®

THERMOGRAPHIC SURVEY INSPECTION SCOPE

Electrical:

- The primary objective is to cover all critical path electrical equipment, high- medium- low-voltage systems, switchgear and equipment control panels
- Incoming power from Utility
- Primary and secondary electrical distribution (down to including 480-V and 120/208-V)
  - Substations
  - Transformers
  - Switchgear
  - Bus run
  - Motor Control Centers
  - Breaker Panels
  - Disconnects
- Emphasis on critical operating/facilities equipment/production machinery
- Critical site motors
- Mainframe, computer room and UPS equipment
- HVAC equipment (if under load)
- Scan other applicable facility/building electrical equipment

Note: This list is not necessarily all-inclusive.

Mechanical – Mechanical equipment often operates in excess of design limits due to misalignment, worn bearings or seals, deteriorated refractory or insulation, etc. in equipment such as:

- Boilers (Refractory, Burner Controls)
- Compressors (if operating)
- Critical pumps / motors
- Fire Pump, Motor/Engine, Controller.
- Refrigeration/HVAC equipment
- Equipment critical to production (as specified)

Note: Mechanical is typically limited to critical equipment.

Ultrasonic Inspection: (High- / Medium- Voltage and Strategic Use Low-Voltage):

- Incoming Utility Services
- Insulation breakdown or air gaps in cable insulation
- Substations, transformers, bushings/insulators, high voltage (as applicable)
- Tracking or ionization on primary switchgear (as applicable)
- Cracked insulators or bushings
- Leaks in sealed transformers

Note: This list is not necessarily all-inclusive.

Visual Inspections:

- Power Panel Labeling- General
- Housekeeping- 3-foot Clearance
- Fireproofing
- Fire Detection
- Clean/Cool/Dry- Main Electrical Switchgear (and proper maintenance)
- Excessive dust/debris
- Transformers (proper operating conditions/maintenance)
- Any substandard conditions, defects, etc. warranting attention
- General electrical preventive maintenance

Note: This list does not include any other visual anomalies which the Thermographer deems appropriate.

DEFINITIONS AND GENERAL NOTES

Temperature Rise: The temperature rise shown for each recommendation is the temperature difference in Celsius (°C) between the object and a reference temperature. When available, a piece of equipment similar to that being scanned is used as the reference. This is usually one phase of a three-phase system. Because of different loading conditions, a comparison like this usually cannot be done for a motor, gear set, etc. In cases when a comparison is not necessary, the actual temperature is shown (such as kiln, boiler, etc.).

Sequence Number In a route based inspection a sequential number is provided indicating that item's relative position in the actual inspection route. This assists in logically locating equipment added during subsequent inspections.

Item Number: Each object scanned is listed and referenced by a unique number.

Recommendation Number Each recommendation is assigned a unique number. The number is a combination of the year and month in which the inspection was completed, along with a sequential number, e.g., 2018-01-1.

Thermogram Number: Each recommendation/finding has two photographs or images of the object. A picture number is assigned for identification purposes. Recommendations are summarized in order of observation (Finding Number) not in order of importance or classification of severity or impact. In some cases, picture numbers, finding number and thermogram number are used interchangeably.

Thermogram: A thermogram is a photograph or image of the thermal image appearing on the infrared display.

Isotherm: Isotherms are white dots electronically superimposed on the display screen and used to illuminate areas of relative identical temperature. Isotherm level readings are used in calculating object surface temperature.

External/Front: A phrase such as "external", "front" or "back", etc., indicate the enclosure or dead-front could not be opened to scan the electrical equipment inside. This is generally due to interlocks that prevent opening the enclosure under load. Only serious or critical recommendations normally generate heat significant enough to be observable and raise the surface temperature of the enclosure.

Bus Ducts/ Bus Plugs: When using infrared scanning to check bus ducts and bus plugs, it should be noted that the temperature-rise given is the surface temperature on the outside of the bus duct at a distance from the target. The actual temperature inside is approximately four times greater.

Airborne Ultrasound: Electrical anomalies occurring within high-, medium- and low-voltage electrical systems can produce short wavelength acoustic sounds inaudible to the human ear (20 – 100 kHz). Such conditions can indicate arcing, tracking, corona, partial discharge, mechanical looseness and other problems within electrical equipment and made audible with the use of ultrasonic technology. Strategic utilization of this non-destructive examination and predictive maintenance equipment is included within your GRC Infrared Thermographic Survey.

Global Risk Consultants®

SEVERITY CODE DESCRIPTION

Classification	Symbol	Temp. Rise	Comments
Minor Problem	*	1° - 9° C	Repair as part of regular maintenance; little probability of physical damage.
Intermediate Problem	**	10° - 34° C	Repair in the near future. Monitor load and watch for changes. Inspect for physical damage.
Serious Problem	***	35° - 74° C	Repair in the immediate future (1-2 days). Inspect the surrounding components for probable damage.
Critical Problem	****	75° C or greater	Repair immediately (overtime). Inspect surrounding components. Repair while Infrared Camera is still available to assess/inspect after correction.
Notification Of Risk Exposure	FYI	FOR YOUR INFORMATION	Observed condition poses a risk to the integrity of the facilities electrical, mechanical, and building operating systems.

In this report, Recommendations are assigned a Severity Code to assist the Client with prioritizing the correction of the detected anomalies. Temperature-rise is not the sole determining factor in this designation. Spot maximum temperature may also be another consideration, along with several other operating or loading conditions and other factors. The Thermographic Consultant may increase the severity of a Recommendation to reflect his/her consideration of the potential losses that the described condition could initiate, importance of the asset, observed damages, etc. The FYI – “For Your Information” Severity Code is used to describe conditions observed by the Thermographic Consultant during his/her inspection that are germane to the preservation of the company’s assets. These conditions are reported in this format for clarity, reference, on-site discussion and documentation.

The Severity Ratings given to the mechanical, refractory, heat transfer and friction point anomalies in this report are predicated on the history of the equipment, it’s nameplated or reference specifications, comparative criteria and experience.

Determination of when, how or who should correct the conditions described in this report remain the sole responsibility of qualified personnel at the client. Please refer to our disclaimer statement.

Global Risk Consultants®

IMPACT CODE DESCRIPTION

Rating	Description	Comments
Low	< 10 K and/or 1 day	Anticipated probable loss of less than $10,000 and/or business interruption of less than 1 day, as reported to the consultant by facility contact or representative.
Moderate	> 10 < 100 K and/or > 1 day< 1 week	Anticipated probable loss of between $10,000 - $100,000 and/or business interruption between 1 and 7 days, as reported to the consultant by facility contact or representative.
High	> $100 K and/or 1 week	Anticipated probable loss of over $100,000 and/or business interruption over 1 week, as reported to the consultant by facility contact or representative.

STATUS CODE DESCRIPTION

Status	Description
Outstanding	As new recommendations are entered, they are set with an "Outstanding" status until there is a facility response and the recommendation is completed.
Rptd. Completed	Upon receipt of plant responses identifying their "Intended Action" as Completed, the recommendation status will be set to "Rpt Completed".
Conf. Completed	After re-inspection is completed, the GRC engineer will confirm that the recommendation is completed.
In Progress	Indicates that the facility has reviewed the recommendation and the action plan is in progress, but not yet completed.
Removed	Is removed as a result of the associated piece of equipment no longer being in operation or removed from service.

HISTORY CODE DESCRIPTION

Status	Description
New	As new recommendations are entered, they are designation as New.
PFR	Previous Finding Repeated denotes a repeat fault from most recent inspection.
PFC	Previous Finding Corrected denotes a corrected fault from most recent inspection.
NIS/NOP	Denotes that the equipment associated with this fault was Not Inspected or Not In Operation at the time of the inspection.

Global Risk Consultants®

Equipment Summary Report

Client: Cargill Incorporated
Division: Starches and Sweeteners
Facility: Research & Development
Location: Vilvoorde, Belgium

File No.: 495.1518
Client No.: 10779
Report ID: 376116
Date Completed: 07-Apr-23

Seq. No.	Status	Equipment	Location	Equip. ID
1	Inspected	HV Switches Cell 1 (SF-6, US Only)	Incoming Substation	121
2	Inspected	HV Switches Cell 2 (SF-6, US Only)	Incoming Substation	173
3	Inspected	HV Switches Cell 3 (SF-6, US Only)	Incoming Substation	174
4	Inspected	HV Switches Cell 4 (SF-6, US Only)	Incoming Substation	175
5	Inspected	HV Switches Cell 5 (SF-6, US Only)	Incoming Substation	122
6	Inspected	HVP 1-1 To 1-4	Incoming Substation	123
7	Fault Info	Electrical	General	171

Find No.:	**2**	**Equipment:**	Panel filters
Rec. No.:	**2023-04-2**	**Specific Item Detail:**	Abnormal visual condition re: dust/dirt contamination.
Severity:	FYI	**Probable Cause:**	Dirty filters blocking the airflow.
History:	New	**Recommendation:**	Filters should be cleaned or replaced frequently to avoid high temperatures in the panels.
Impact:	Low	**Redundancy Spare:**	Spare - Inline Redundancy

8	Inspected	LV Switch X2 400A	B-1	10
9	Inspected	VB.B01 Panel	B-1	11
10	Inspected	A414-2 Panel	B-1	82
11	Inspected	B430-4 Panel	B-1	83
12	Inspected	VB.B11 Panel	B-1	12
13	Inspected	VB.B21 Panel	B-1	13
14	Inspected	Panel	B-1	14
15	Inspected	Technisch Verdiep Panel	B-1	15
16	Inspected	Computer Panel	B-1	16
17	Inspected	Lift "B" Panel	B-1	17
18	Inspected	Vb. CO1 Panel	B-1	18
19	Inspected	Fermantatie Panel	B-1	19
20	Inspected	Bord O.P.P	B-1	20
21	Inspected	Bord B100-11A	B-1	21
22	Inspected	Bord B100-11B	B-1	22
23	Inspected	B100-12 Distribution Board	B-1	84
24	Inspected	Fire Evacuation Alarm System C.P.	B-1	152
25	Inspected	Fire Evacuation Alarm System Batteries	B-1	153
26	Inspected	VBB.01 Switch Panel (2)	B050	9

This report does not purport to set forth all hazards nor to indicate that other hazards do not exist. However, we urge that recommendations presented, if any, be given serious consideration. We would be pleased to discuss alternative solutions.

Global Risk Consultants®

Equipment Summary Report

Client: Cargill Incorporated
Division: Starches and Sweeteners
Facility: Research & Development
Location: Vilvoorde, Belgium

File No.: 495.1518
Client No.: 10779
Report ID: 376116
Date Completed: 07-Apr-23

Seq. No.	Status	Equipment	Location	Equip. ID
27	Inspected	LV-B0-51-001-22	B0	178
28	Inspected	LV-B0-51-002-12	B0	179
29	Inspected	B136-1 Computer Supply	B136, Server Room	24
30	Inspected	Servers	B136, Server Room	172
31	Inspected	UPS1	B136, Server Room	176
32	Inspected	B111-1 Bord CTHR	B111	23
32	Inspected	UPS2	B136, Server Room	177
33	Inspected	VBB.1.1 Distribution Panel	B150	25
34	Inspected	VBB.2.1 Distribution Panel	B250	26
35	Inspected	VBB.3.1 Distribution Panel	B350	27
36	Inspected	C335-1 Bord Stookplaats	C335	37
37	Fault Info	B430-1 HVAC Control Panel, LV_B+4-30-001-13	B430	28

Find No.:	1	**Equipment:**	39Q13
Rec. No.:	2023-04-1	**Specific Item Detail:**	Abnormal thermal condition re: the breaker.
Severity:	***	**Probable Cause:**	Overloading suspected.
History:	New	**Recommendation:**	Check load versus rated capacity. Ensure proper cooling.
Impact:	Low	**Redundancy Spare:**	Spare - Inline Redundancy

38	Inspected	B430-2 AC Supply Panel	B430	85
39	Inspected	Fume Extraction Fans (~40)	B430	151
41	Inspected	VBC.0.1 Supply Panel	C030 Confectionery	30
42	NIS/NOP	Busbar	C030 Confectionery	148
43	Inspected	VBC.2.1/C103-1 Supply Panel	C130	31
44	Inspected	C130-2 Bord Collector	C130	32
45	Inspected	C130-4 Bord PG5EG5	C130	33
46	Inspected	Bord C130-3 PG4-EG4	C130	34
47	Inspected	C130-5 Freezer Control Panel	C130	35
48	Inspected	Freezer Door	C130	149
49	Inspected	VBC.3.1 C230-1SUPPLY Panel	C230	36
50	Inspected	Busbar	C230	150
51	Inspected	A008-1 Main Supply	A008 Toilets	1
52	Inspected	A001-1 Distribution Panel	A001, Reception	88

This report does not purport to set forth all hazards nor to indicate that other hazards do not exist. However, we urge that recommendations presented, if any, be given serious consideration. We would be pleased to discuss alternative solutions.

Global Risk Consultants®

Equipment Summary Report

Client: Cargill Incorporated
Division: Starches and Sweeteners
Facility: Research & Development
Location: Vilvoorde, Belgium

File No.: 495.1518
Client No.: 10779
Report ID: 376116
Date Completed: 07-Apr-23

Seq. No.	Status	Equipment	Location	Equip. ID
53	Inspected	Bord A108-1	A108	2
54	Inspected	Bord A208-1	A208	3
55	Inspected	Bord A308-1	A308	4
56	Inspected	A414-1 HVAC Control Panel	A414	7
57	Inspected	A414-3 Distribution Panel	A414	86
58	Inspected	A414-2 Distribution Panel	A414	87
59	Inspected	A404-1 Liftkamer	A404	5
60	Inspected	R209-1 Distribution Panel	R209	90
61	Inspected	R003-1	R003	89
62	Removed	F002-1 Verdeelbord OPP Plant	F002	38
63	Removed	F102-1 Eb Stookplaats O-P-P	F102 Heating Room	39
64	Inspected	Verdeelbord Meat	D Pilot Meat and Pharma	168
65	Inspected	Verdeelbord Pharma	D Pilot Meat and Pharma	169
66	Inspected	Verdeelbord Meat 2	D Pilot Meat and Pharma	170
67	Inspected	J050-1 Distribution Panel (1-2)	J050	40
68	Inspected	J050-1-1 Distribution Panel (2)	J050	180
69	Inspected	Busbar 250	J050	147
70	NIS/NOP	Compressor I0021	J033	145
71	NIS/NOP	Bord J1.30	J1	181
72	Inspected	Bord LV_J+1_30_U549_001_13	J1	184
73	Inspected	Bord LV_J+2_30_U547_001_13	J2	185
74	Inspected	Busbar	J2	182
75	Inspected	Bord LV_J+3.36_001_13	J3	183
76	NIS/NOP	Busbar	J304	42
77	Inspected	K.0.80-01, Lighting Control Panel	K0, Under Stairs	100
78	Inspected	K.1.19.01, HVAC	K1 Technical Room	101
79	Inspected	TC AIR-SB02	K1 Technical Room	102
80	Inspected	Weiss Panel	K1 Technical Room	103
81	Inspected	Compressor Koelkamer 1-2	K1 Technical Room	143

This report does not purport to set forth all hazards nor to indicate that other hazards do not exist. However, we urge that recommendations presented, if any, be given serious consideration. We would be pleased to discuss alternative solutions.

Equipment Summary Report

Client: Cargill Incorporated
Division: Starches and Sweeteners
Facility: Research & Development
Location: Vilvoorde, Belgium

File No.: 495.1518
Client No.: 10779
Report ID: 376116
Date Completed: 07-Apr-23

Seq. No.	Status	Equipment	Location	Equip. ID
82	Inspected	Compressor Diepvries	K1 Technical Room	144
83	Inspected	K.2.28-01, Office	K2.10	104
84	Inspected	K.3.01-01, Labo	K3.12	105
85	Inspected	Bms/Noodverlichting	P3, K3.07 Fermentation	106
86	NIS/NOP	Panel Hosokawa Dryer	P3	166
87	Inspected	Teblick Panels	P3	107
88	Inspected	AHU Panels (Panels 1-2)	P3	108
89	Inspected	AHU Control Panel Small	P3	141
90	NIS/NOP	Oil Heater 10803	P3	142
91	Inspected	TD HVAC	P3	167
92	Inspected	P.2.50-1 Panels 1-3	P3	109
93	Inspected	P.2.50-2 Panels 1-2	P3	110

Find No.:	**502**	**Equipment:**	Connectionblock X4-57
Rec. No.:	**2022-04-2**	**Specific Item Detail:**	Abnormal thermal condition re: the connection(s).
Severity:	******	**Probable Cause:**	Two cable in one connection
History:	**PFC**	**Recommendation:**	This type of connectionblock only allows to fit one wire. Re-design or use a double crimp lug.
Impact:	**Low**	**Redundancy Spare:**	Spare - Inline Redundancy

94	Inspected	Voedingsbord 140203HX003	P3	139
95	Inspected	P3.50.01	P3	140
96	Inspected	140204-2	P2 Technical Corridor	111
97	Inspected	140203-1	P2 Technical Corridor	134
98	Inspected	140204-1	P2 Technical Corridor	112
99	NIS/NOP	CM20 140201-1	P2 Technical Corridor	135
100	Inspected	Panel	P2 Technical Corridor	136
101	NIS/NOP	Extruder U0049 MCC1	P2 Technical Corridor	137
102	Inspected	CB20 141201-1	P2 Technical Corridor	113
103	Inspected	Door Control	P2 Corridor, P2.92	138
104	Removed	Bord EB3	P2.20	114
105	Removed	Bord Labo P104-1	P2.21	132
106	Inspected	Busbar MDP1/29Q1	P2.20	133
107	Not Inspecte	Compressor 1.0701 (Interlocked)	P3, Compressor Room	99

This report does not purport to set forth all hazards nor to indicate that other hazards do not exist. However, we urge that recommendations presented, if any, be given serious consideration. We would be pleased to discuss alternative solutions.

Equipment Summary Report

Client: Cargill Incorporated
Division: Starches and Sweeteners
Facility: Research & Development
Location: Vilvoorde, Belgium

File No.: 495.1518
Client No.: 10779
Report ID: 376116
Date Completed: 07-Apr-23

Seq. No.	Status	Equipment	Location	Equip. ID
108	Not Inspecte	Compressor 1.0702 (Interlocked)	P3, Compressor Room	129
109	Inspected	Filter 1.0703	P3, Compressor Room	130
110	Inspected	Control Panel 3X400-V	P3, Compressor Room	131
111	Removed	Panel P105-2	P3, Extraction Room	165
112	Inspected	Panel - Solar Panels	P3, Extraction Room	187
113	Inspected	MDP-2 Panel 1-5	P3, LV Room	92
114	Inspected	LV Busbar (X3)	P3, LV Room	126
115	Inspected	MDP-3 (1-2)	P3, LV Room	93
116	Inspected	MDP-1 (1/1-1/6)	P3, LV Room	94
117	Inspected	Cos Phi (Comar) 90-KVAR	P3, LV Room	95
118	Inspected	Cos Phi (Comar) 210-KVAR	P3, LV Room	127
119	Inspected	Trafo 1000 KVA (Double Winding 2-TAPS) (Dry) [IR/US]	P3 (HV Room)	96
120	Inspected	Trafo 1600 KVA (Dry) [IR/US]	P3 (HV Room)	97
121	Inspected	HV Switches (Cells 1-3) (US Only) Vacuum	P3 (HV Room)	98
122	Inspected	UPS (EPA) 1	P3 (HV Room)	124
123	Inspected	UPS (EPA) 2	P3 (HV Room)	125
124	Inspected	TC Cold	P1	115
125	NIS/NOP	Daikin Chiller	P1	120
126	Inspected	2 Motors	P1	163
127	Inspected	P1.80 Panel	P1	164
128	Inspected	TC Chaufferie HEAT-SB01	P0, Steam Generator Room	116
129	Inspected	P.0.82-1	P0, Steam Generator Room	117
130	NIS/NOP	I0700 Steam Generator	P0, Steam Generator Room	118
131	Inspected	I0782 Steam Generator	P0, Steam Generator Room	160
132	Inspected	Operating Motors	P0, Steam Generator Room	161
133	Inspected	I0704	P0, Water Treatment	119

Find No.: 501
Rec. No.: 2022-04-1
Severity: ****
History: PFC
Impact: Low

Equipment: Cooling Fan
Specific Item Detail: Abnormal thermal condition re: the fan.
Probable Cause: Fan/component deteriorated.
Recommendation: Replace fan or component as needed.
Redundancy Spare: No spare

This report does not purport to set forth all hazards nor to indicate that other hazards do not exist. However, we urge that recommendations presented, if any, be given serious consideration. We would be pleased to discuss alternative solutions.

Equipment Summary Report

Client: Cargill Incorporated
Division: Starches and Sweeteners
Facility: Research & Development
Location: Vilvoorde, Belgium

File No.: 495.1518
Client No.: 10779
Report ID: 376116
Date Completed: 07-Apr-23

Seq. No.	Status	Equipment	Location	Equip. ID
134	Inspected	Operating Motors	P0, Water Treatment	162
135	Inspected	Q106-1 ATELIER-1	Q006	60
136	Inspected	Q006	Q006	158
137	Inspected	Q014 (X2)	Q14, Workshop Store	159
138	Inspected	Q106-2 Atelier 1	Q106, Mezz	61
139	Inspected	Q007-2 (X2)	Q007 Workshop Area	156
140	Inspected	Q008-1 Stookplaats 1	Q008	67
141	Inspected	Q008-2 Distribution Panel	Q008	155
142	Inspected	Bord I0035, SIHI Vacuumpompen	Q008	91
143	Inspected	Q001-1 Magazijn 1	Q001	59
144	Inspected	Workstation Cables/Plugs	Q001, Warehouse Office	154

This report does not purport to set forth all hazards nor to indicate that other hazards do not exist. However, we urge that recommendations presented, if any, be given serious consideration. We would be pleased to discuss alternative solutions.

Global Risk Consultants®

Recommendation Summary Report

Client: Cargill Incorporated
Division: Starches and Sweeteners
Facility: Research & Development
Location: Vilvoorde, Belgium

File No.: 495.1518
Client No.: 10779
Report ID: 376116
Date Completed: 07-Apr-23

Finding/ Rec No./ History	Seq. No.	Equipment	Location	Specific Item	Severity/ Impact/ Status
1 2023-04-1 New	37	B430-1 HVAC Control Panel, LV_B+4-30-001-13, 39Q13	B430	Abnormal thermal condition re: the breaker.	*** Low Outstanding
2 2023-04-2 New	7	Electrical, Panel filters	General	Abnormal visual condition re: dust/dirt contamination.	FYI Low Outstanding
501 2022-04-1 **PFC**	133	I0704, Cooling Fan	P0, Water Treatment	Abnormal thermal condition re: the fan.	**** Low Conf Completed
502 2022-04-2 **PFC**	93	P.2.50-2 Panels 1-2, Connectionblock X4-57	P3	Abnormal thermal condition re: the connection(s).	** Low Conf Completed

This report does not purport to set forth all hazards nor to indicate that other hazards do not exist. However, we urge that recommendations presented, if any, be given serious consideration. We would be pleased to discuss alternative solutions.

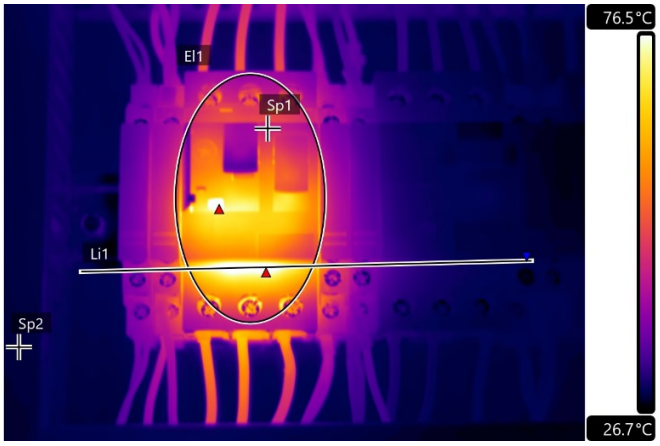
Draft Copy

Global Risk Consultants®

Cargill R&D	Vilvoorde, Belgium	WKID: 376116	File No: 495.1518

Finding No: 1 Rec No: 2023-04-1 Seq. No: 37	Temp Rise: 50°C Severity: *** . Impact: Low .

Location:	B430
Equipment:	B430-1 HVAC Control Panel, LV_B+4-30-001-13, 39Q13
Specific Item:	Abnormal thermal condition re: the breaker.







Date:	4/7/2023 8:00:53 AM
Camera Model:	FLIR T660
Camera Serial Number:	55912512
Maximum Temp:	97.6 °C
Reference: Sp1	48.1 °C
Ambient: Sp2	28.6 °C
Max Line:	79.5 °C

Probable Cause:	Overloading suspected.
Recommendation:	Check load versus rated capacity. Ensure proper cooling.
Comments:	Check breaker size, expected to be loaded 100%. investigate to increase size. Breaker 6-10A, set at 10A.

This report does not purport to set forth all hazards nor to indicate that other hazards do not exist. However, we urge that recommendations presented, if any, be given serious consideration. We would be pleased to discuss alternative solutions.

Global Risk Consultants®

Cargill R&D	Vilvoorde, Belgium	WKID: 376116	File No: 495.1518

Finding No: 2 Rec No: 2023-04-2 Seq. No: 7	Temp Rise: N/A ***Severity:*** ***FYI*** Impact: Low .

Location:	General
Equipment:	Electrical, Panel filters
Specific Item:	Abnormal visual condition re: dust/dirt contamination.





Probable Cause:	Dirty filters blocking the airflow.
Recommendation:	Filters should be cleaned or replaced frequently to avoid high temperatures in the panels.
Comments:	Images above are just 2 examples, many more dirty filters were found.

This report does not purport to set forth all hazards nor to indicate that other hazards do not exist. However, we urge that recommendations presented, if any, be given serious consideration. We would be pleased to discuss alternative solutions.